

FOR VPES AND ENGINEERING DIRECTORS AT 200 TO 500 ENGINEERS

The AI ROI Calibration Matrix

A scorecard for separating AI activity metrics from AI leverage metrics, with the calibration questions that distinguish the two and the workflow segmentation most orgs skip.

01 / THE MISFRAME

You are instrumenting AI like a software license. That is the wrong altitude.

Most orgs measure AI by inputs: tokens consumed, PRs closed, lines of code generated, percentage of commits with AI involvement. These metrics tell you the tool is running. They do not tell you whether it is producing leverage.

Jellyfish tracked 20,000 engineers across 300 companies. 25% speed-up in coding time. 12% increase in PR throughput.

Both numbers are inputs. Neither one answers the question a board or a CFO is actually asking: did engineering capacity increase, or did engineers just get faster at producing the same shaped output with worse downstream margin?

THE REFRAME

Activity metrics ask "is AI running." Leverage metrics ask "how much meaningful output per unit of human judgment applied." Until your instrumentation answers the second question, you cannot run an honest ROI calculation.

02 / THE CALIBRATION MATRIX

How to classify the metrics already on your dashboard

METRIC	CLASSIFICATION	WHAT IT ACTUALLY TELLS YOU
AI coding time saved	Activity / input	Tool adoption signal. Not a leverage signal. Engineers can be 25% faster and still ship fewer net features.
PR throughput delta	Activity / input	Quantity of shipped units. Says nothing about quality, downstream rework, or production health.
Tokens or AI cost per engineer	Cost / input	Operational unit economics of the tool. Required for budget. Not a leverage measure.
Feature shipped to production holding SLO on first deploy	Leverage / output	The unit of meaningful output. Measures actual shipped value, net of rework, regression, hotfix.
Reverted commits within 7 days, AI-attributed	Leverage / quality	Captures the cost of AI-generated code that compiled but misfired. Catches the failure mode activity metrics hide.
MTTR on AI-authored modules	Leverage / risk	Surfaces the comprehension cost of AI authorship. Usually 30 to 60% higher than human-authored modules at 200+ engineers.
Net features shipped per engineer per quarter	Leverage / output	The "meaningful output per unit of judgment" denominator. The single most useful number a VPE can present to the board.

Map every metric your team currently reports against this table. The metrics on the activity rows are not wrong, they are just at the wrong altitude for an ROI conversation. The metrics on the leverage rows are what your CFO is actually asking about.

03 / WORKFLOW SEGMENTATION

Average leverage hides where AI is actually working and where it is silently degrading

Once you have leverage metrics, segment by workflow type before you draw conclusions. AI produces dramatically different leverage on different categories of work, and an average hides where the actual gain or loss is concentrating. Three segments cover most engineering work at 200 to 500 engineer scale:

SEGMENT 01

Repetitive boilerplate

CRUD endpoints, scaffolding, test fixtures, internal admin tooling, well-trodden refactors.

Expected leverage: HIGH

SEGMENT 02

Novel logic in a known system

New features inside familiar architecture, integration work, parameter tuning, custom pipelines.

Expected leverage: MEDIUM

SEGMENT 03

Architectural judgment

System design, performance critical paths, cross-team contracts, novel data structures, incident response.

Expected leverage: NEGATIVE

The Segment 3 work is where AI activity metrics most commonly mislead. AI-generated code in this category produces plausible-looking output that ships fast and fails six months later when the original reasoning was never internalized by a human. **Activity metrics call this a win. Leverage metrics call it deferred cost.**

04 / CALIBRATION CHECKLIST

30-minute audit your engineering leadership team can run this week

- ☐ **List every AI metric your org currently reports.** Map each one to the Calibration Matrix as activity, cost, or leverage. Count how many fall into each bucket.
- ☐ **Audit your last 4 board or leadership decks.** Identify which AI numbers were presented. If more than 70% were activity metrics, the ROI conversation has been theater.
- ☐ **Define your unit of meaningful output.** For most 200 to 500 engineer orgs the right unit is "feature shipped to production that holds its SLO on first deploy." Confirm with your VP Product and your SRE lead.
- ☐ **Segment by workflow type for one team.** Pick a single team. Tag their last 60 days of work into the three segments. Calculate leverage per segment, not averaged.
- ☐ **Instrument reverted-within-7-days for AI-attributed commits.** This is the cheapest leverage metric to install and the one that surfaces silent degradation fastest.
- ☐ **Compare MTTR on modules above and below 40% AI authorship.** If the AI-heavy modules have meaningfully higher MTTR, you have a comprehension debt signal worth governance attention.
- ☐ **Replace one activity metric with one leverage metric in your next board update.** Pick the most consequential swap. Note what the conversation becomes when the altitude changes.

05 / DISCUSSION QUESTIONS

For your next engineering leadership team meeting

1. What is the single piece of evidence that would convince our board that AI tooling is producing leverage, not just speed? Do we currently produce that evidence?
2. Which of the three workflow segments is our team using AI on most heavily? Is the leverage we are seeing concentrated where we expect it, or distributed in a way that suggests we are over-using AI on Segment 3?
3. If our headcount-for-AI cost swap was wrong, what is the earliest signal we would see, and how many quarters before it becomes a board conversation?

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